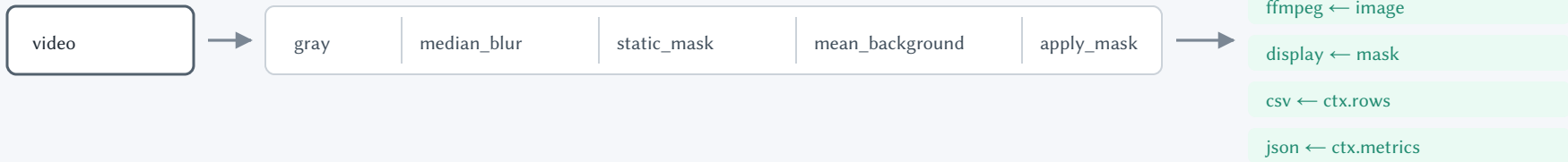


source → stages → sinks, once per frame



Ctx — one per frame, handed down the chain

image	the working array — a stage rebinds it, never mutates it
source	the original BGR frame, untouched by anything
index	the frame number
store	side channel within one frame: `mask`, `heat`, `boxes`
taps	a named stage's output, kept for a sink to pick from
metrics	per-frame scalars — what the json sink writes
rows	per-object detail — what the csv and crops sinks write

Two kinds of state, deliberately apart: what survives the frame lives on the stage instance, what does not lives in `store`.

Why a side channel

A plain `ndarray` → `ndarray` chain cannot express what the gas recipe needs:

`mean\_background` accumulates the masked frame but subtracts from the unmasked one.

`static\_mask` writes the mask, two later stages read it, and the frame carries on unchanged.

Nothing branches on runtime state; there is no state machine.